C.6. Inverse Function Theorem. Let $U \subset \mathbb{R}^n$ be an open set and suppose $\mathbf{f} : U \to \mathbb{R}^n$ is C^1, $\mathbf{f} = (f^1, \dots, f^n)$. Assume $x_0 \in U,\ z_0 = \mathbf{f}(x_0)$.

Remember from §A.4 that we write

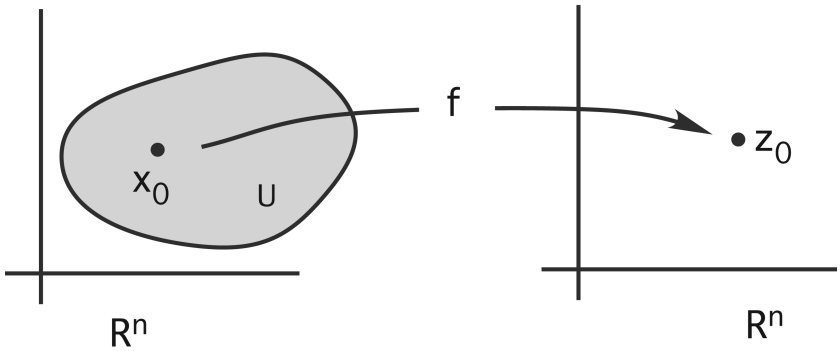
$$D\mathbf{f} = \begin{pmatrix} f^1_{x_1} & \cdots & f^1_{x_n} \\ & \ddots & \\ f^n_{x_1} & \cdots & f^n_{x_n} \end{pmatrix}_{n\times n} = \textit{gradient matrix} \text{ of } \mathbf{f}.$$

DEFINITION.

$$J\mathbf{f} = \textit{Jacobian of } \mathbf{f} = |\det D\mathbf{f}| = \left|\frac{\partial(f^1, \dots, f^n)}{\partial(x_1, \dots, x_n)}\right|.$$

THEOREM 8 (Inverse Function Theorem)**.** *Assume* $\mathbf{f} \in C^1(U; \mathbb{R}^n)$ *and*

$$J\mathbf{f}(x_0) \neq 0.$$



Then there exist an open set $V \subset U$, with $x_0 \in V$, and an open set $W \subset \mathbb{R}^n$, with $z_0 \in W$, such that

(i) *the mapping*

$$\mathbf{f} : V \to W$$

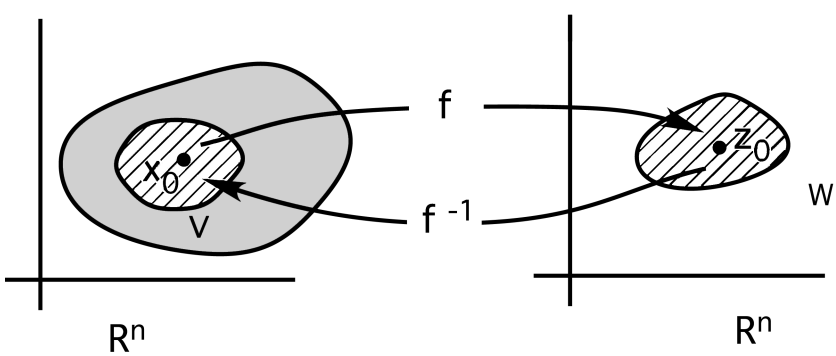
is one-to-one and onto and

(ii) *the inverse function*

$$\mathbf{f}^{-1} : W \to V$$

is C^1.

(iii) *If $\mathbf{f} \in C^k$, then $\mathbf{f}^{-1} \in C^k$ $(k = 2, \dots)$.*



C.7. Implicit Function Theorem. Let n, m be positive integers. We write a typical point in $\mathbb{R}^{n+m}$ as

$$(x, y) = (x_1, \dots, x_n, y_1, \dots, y_m)$$

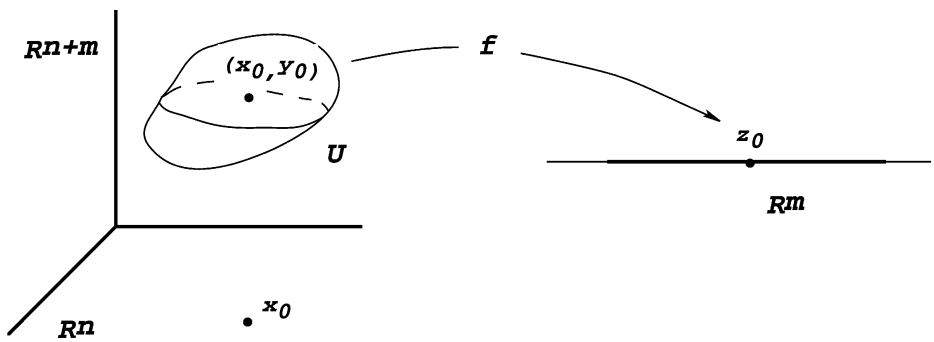
for $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$.

Let $U \subset \mathbb{R}^{n+m}$ be an open set and suppose $\mathbf{f} : U \to \mathbb{R}^m$ is C^1, $\mathbf{f} = (f^1, \dots, f^m)$. Assume $(x_0, y_0) \in U$, $z_0 = \mathbf{f}(x_0, y_0)$.

NOTATION.

$$D\mathbf{f} = \begin{pmatrix} f^1_{x_1} & \cdots & f^1_{x_n} & f^1_{y_1} & \cdots & f^1_{y_m} \\ & \ddots & & & \ddots & \\ f^m_{x_1} & \cdots & f^m_{x_n} & f^m_{y_1} & \cdots & f^m_{y_m} \end{pmatrix}_{m\times(n+m)}$$

$$= (D_x\mathbf{f}, D_y\mathbf{f}) = \textit{gradient matrix of } \mathbf{f}.$$



DEFINITION.

$$J_y\mathbf{f} = |\det D_y\mathbf{f}| = \left|\frac{\partial(f^1, \dots, f^m)}{\partial(y_1, \dots, y_m)}\right|.$$

THEOREM 9 (Implicit Function Theorem). *Assume* $\mathbf{f} \in C^1(U; \mathbb{R}^m)$ *and*

$$J_y\mathbf{f}(x_0, y_0) \neq 0.$$

Then there exists an open set $V \subset U$, *with* $(x_0, y_0) \in V$, *an open set* $W \subset \mathbb{R}^n$, *with* $x_0 \in W$, *and a* C^1 *mapping* $\mathbf{g} : W \to \mathbb{R}^m$ *such that*

(i) $\mathbf{g}(x_0) = y_0$,

(ii) $\mathbf{f}(x, \mathbf{g}(x)) = z_0 \quad (x \in W)$, *and*

(iii) *if* $(x, y) \in V$ *and* $\mathbf{f}(x, y) = z_0$, *then* $y = \mathbf{g}(x)$.

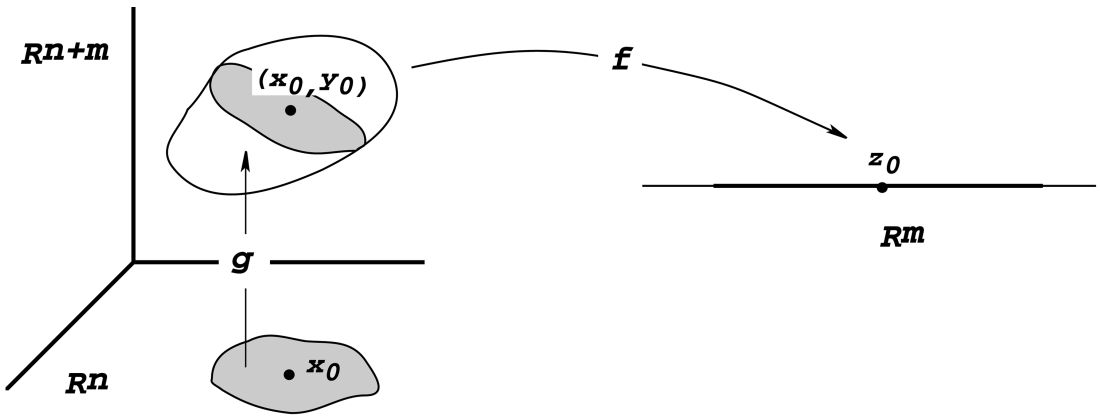
(iv) *If* $\mathbf{f} \in C^k$, *then* $\mathbf{g} \in C^k$ $(k = 2, \dots)$.

The function $\mathbf{g}$ is *implicitly defined* near x_0 by the equation $\mathbf{f}(x, y) = z_0$.

C.8. Uniform convergence. We record here the *Arzela–Ascoli compactness criterion* for uniform convergence:

Suppose that $\{f_k\}_{k=1}^{\infty}$ is a sequence of real-valued functions defined on $\mathbb{R}^n$, such that

$$|f_k(x)| \leq M \quad (k = 1, \dots, \ x \in \mathbb{R}^n)$$



for some constant M and that the functions $\{f_k\}_{k=1}^{\infty}$ are *uniformly equicontinuous*, meaning that for each $\varepsilon > 0$, there exists $\delta > 0$ such that $|x - y| < \delta$ implies $|f_k(x) - f_k(y)| < \varepsilon$, for $x, y \in \mathbb{R}^n$, $k = 1, \ldots$.

Then there exists a subsequence $\{f_{k_j}\}_{j=1}^{\infty} \subseteq \{f_k\}_{k=1}^{\infty}$ and a continuous function f, such that

$$f_{k_j} \to f \quad \text{uniformly on compact subsets of } \mathbb{R}^n.$$